# STAR: A Station-Aware Temporal-Attention Residual Network for Multi-Station PM2.5 Forecasting in Bangladesh

***Abstract*—Accurate $PM_{2.5}$ forecasting is critical for public-health planning in Bangladesh, where particulate pollution routinely exceeds safe limits despite sparse and heterogeneous monitoring stations. A strong recent paradigm combines the Prophet decomposition model with a recurrent network to learn Prophet residuals.However, recurrence is computationally heavier, less interpretable, and station-agnostic. This paper presents STAR, a Station-aware Temporal-Attention Residual network that predicts Prophet residuals from a 14-day feature window using a single Transformer self-attention layer and conditions on the monitoring station through a Feature-wise Linear Modulation (FiLM) layer, allowing station-specific feature adaptation. STAR is evaluated on a nine-year, sixteen-station Bangladesh air-quality dataset (30,095 station-days) using a leakage-safe per-station time split and compared with a reproduced Prophet+LSTM model, three feed-forward residual baselines, and Prophet on a common test subset of 4,867 observed days. All models are evaluated within a unified statistical framework using bootstrap confidence intervals, a Friedman omnibus test, Holm-corrected pairwise Wilcoxon tests and a Nemenyi critical-difference analysis. STAR achieves the lowest mean absolute error (16.71 vs. 17.15 for LSTM) and the best symmetric MAPE, statistically matching the recurrent baseline while significantly outperforming it at the per-sample level ($p < 10^{-5}$, Holm-corrected) and consistently surpassing all feed-forward residual models. These results show that attention with FiLM-based station conditioning can match recurrent models with lower architectural cost and improved interpretability.**

***Index Terms*—PM2.5 forecasting, air-quality prediction, Prophet, self-attention, FiLM, residual learning, statistical model comparison, Bangladesh.**

## I. Introduction

Fine particulate matter (PM2.5) is a major environmental health risk linked to cardiovascular and respiratory disease and premature mortality. Bangladesh records some of the world's highest PM2.5 levels, with winter peaks driven by trans-boundary transport, brick kilns, vehicular emissions and stable atmospheric conditions. Reliable short-horizon PM2.5 forecasting can support timely public-health interventions but sparse monitoring and strong inter-station variability make accurate forecasting challenging.

Classical decomposition models such as Prophet [1] effectively capture trend and multi-scale seasonality but leave temporally structured residuals. A common two-stage approach uses Prophet for trend and seasonality, while a neural network predicts the residual. The strongest reported variant combines Prophet with an LSTM [2]. However, recurrence is sequential, less interpretable and overlooks station-specific differences.

This paper investigates whether the recurrent residual learner can be matched or surpassed by a lighter, more transparent and explicitly station-aware design. The proposed STAR model replaces recurrence with a single Transformer self-attention layer [3] over the same 14-day feature window, enabling parallel temporal interactions with interpretable attention weights. STAR further incorporates a FiLM layer [4], where a station embedding generates per-channel scale and shift parameters to adapt the pooled temporal representation while keeping temporal parameters shared.

A second focus is methodological rigour. Prophet, the reproduced Prophet+LSTM, three feed-forward residual baselines and STAR are evaluated on a common test subset using a unified statistical framework comprising bootstrap confidence intervals, a Friedman omnibus test, Holm-corrected pairwise Wilcoxon tests and a Nemenyi critical-difference analysis with the sixteen stations as blocks.

The main contributions of this study are:

1) STAR (Station-Aware Transformer for Residual $PM_{2.5}$ Forecasting) introduces a lightweight Transformer-based residual-learning framework that combines self-attention over a 14-day temporal window with FiLM-based station conditioning to capture temporal dependencies while incorporating explicit station awareness through per-channel feature modulation.
2) A leakage-safe Residual-AR feature augmentation method appends lagged Prophet residuals together with rolling statistical features to every residual learner, providing a unified feature representation while preventing information leakage.
3) A unified statistical evaluation framework is established for multi-station $PM_{2.5}$ forecasting using bootstrap confidence intervals, Friedman ranking, and Holm-corrected pairwise Wilcoxon tests, enabling rigorous and fair comparison across sixteen monitoring stations in Bangladesh.

The remainder of this paper is organized as follows. Section II reviews the related work. Section III presents the proposed methodology. Section IV reports the experimental results and statistical analysis. Finally, Section V concludes the paper and discusses future research directions.

## II. Related Work

Prophet [1] models trend and multi-scale seasonality with minimal tuning and tolerates missing data, yet leaves short-term residual dynamics unmodeled. Milli [2] combined Prophet with an LSTM for Beijing PM2.5 forecasting and reported strong improvements over deep-learning baselines. However, the study lacked station-level blocking and multiplicity-controlled pairwise testing, a gap addressed in this work. The ARIMA family [5] provides a strong foundation for linear time-series forecasting but performs poorly on nonlinear pollutant dynamics.

LSTM [6] and GRU [7] have become standard architectures for modeling temporal dependencies in air-quality forecasting. Several studies demonstrated their effectiveness through LSTM-GRU hybrids [8], LSTM-FC models [9], recurrent forecasting frameworks [10], [11], and CNN-LSTM architectures [12]. Xiao et al. [13] further incorporated station distance and pollution similarity before an LSTM stage, providing an early form of station-aware modeling that STAR generalizes through learned FiLM conditioning. Later studies also explored graph convolutional LSTM models [14] and CNN-BiLSTM networks [15] for capturing spatial and temporal dependencies.

The Transformer [3] introduced self-attention for parallel sequence modeling with interpretable attention maps, while FiLM [4] enables feature modulation through learned conditioning. Cui et al. [23] demonstrated the effectiveness of Transformers for long-range $PM_{2.5}$ prediction. Recent studies further explored sparse attention [25], spatio-temporal attention [26], Transformer-BiLSTM hybrids [27], seasonal-trend decomposition [28], multi-granularity fusion [29] and the Temporal Fusion Transformer [30]. Wu et al. [24] highlighted the need for more rigorous benchmarking, while Zhou et al. [31] noted that station-aware conditioning remains underexplored, a gap addressed by STAR through FiLM conditioning.

Early studies explored both neural and statistical approaches for air-quality forecasting. Ordieres et al. [16], Barai et al. [17], and Corani [18] evaluated MLP, radial-basis-function, recurrent, self-organising-map, feed-forward and lazy-learning models. Russo et al. [19] improved neural-network robustness using stochastic inputs. While Hu et al. [20], Yang et al. [21] and Fu [22] proposed statistical models based on aerosol retrievals, wavelet decomposition and online multivariate regression for $PM_{2.5}$ forecasting.

Shahriar et al. [32] compared ARIMA-ANN, ARIMA-SVM, decision-tree and CatBoost hybrids for $PM_{2.5}$ forecasting in Bangladesh but evaluated stations independently without station-aware modeling or multiplicity-controlled comparison. Rahman et al. [33] developed AirNet for real-time air-quality forecasting, while Mandal and Sen [34] proposed a bidirectional GRU for AQI prediction. Islam et al. [35] compared the Temporal Fusion Transformer with DeepAR, GAN and CNN models across four Bangladeshi cities. But none employed the station-blocked, multiplicity-controlled statistical evaluation adopted in this work.

Demšar [36] established statistical comparison of multiple predictors using the Friedman test [37], the Nemenyi post-hoc test and the critical-difference diagram, while the Wilcoxon test [38] with Holm's correction [39] controls the family-wise error rate across pairwise comparisons. These methods are rarely applied together in air-quality forecasting. This work adopts them using the sixteen monitoring stations as statistical blocks.

## III. Methodology

The final forecast is the sum of the Prophet baseline and a learned residual:

$$\hat{y}_t = \hat{y}_t^P + \hat{r}_t, \quad r_t = y_t - \hat{y}_t^P, \tag{1}$$

where $\hat{y}_t^P$ is Prophet's prediction and $\hat{r}_t$ is the residual predicted by a neural learner. All residual learners share the same Prophet baseline, feature set and training protocol, ensuring that performance differences are attributable solely to the residual learning module.

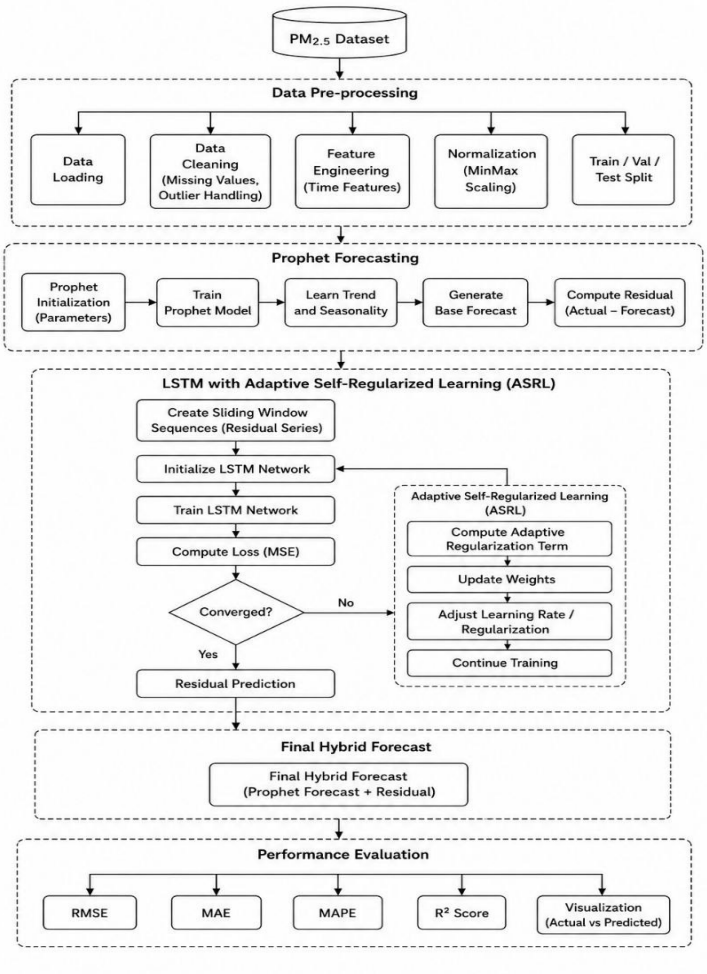



Fig. 1. Overview of the proposed STAR framework for $PM_{2.5}$ forecasting.

Figure 1 presents the overall workflow of the proposed STAR framework from Prophet-based forecasting and residual learning to final model evaluation.

### A. Dataset and Preprocessing

A Bangladesh air-quality dataset comprising sixteen monitoring stations and 662,651 hourly observations collected between November 2012 and December 2021 is used in this study. Table I summarizes the key characteristics of the dataset. $PM_{2.5}$ is considered the prediction target, while $PM_{10}$, $SO_2$, NO, $NO_2$, $NO_x$, CO, $O_3$, wind speed, wind direction, temperature, relative humidity, solar radiation, barometric pressure, and rainfall are used as predictor variables. Duplicate observations are removed during preprocessing, missing values are handled through station-wise interpolation, and wind direction is transformed into sine and cosine components to preserve its circular nature. The hourly observations are then aggregated into daily station-level records, which are subsequently used for feature engineering, model training, and evaluation.

TABLE I
OVERVIEW OF THE BANGLADESH $PM_{2.5}$ DATASET

| Attribute | Description |
|---|---|
| Dataset | Bangladesh Air Quality Dataset |
| Study period | Nov. 2012 – Dec. 2021 |
| Monitoring stations | 16 |
| Hourly records | 662,651 |
| Prediction target | $PM_{2.5}$ |
| Predictor variables | $PM_{10}$, $SO_2$, NO, $NO_2$, $NO_x$, CO, $O_3$, Wind Speed, Wind Direction, Temperature, RH, Solar Radiation, Barometric Pressure, Rainfall |
| Temporal resolution | Hourly (aggregated to daily) |

### B. Split and Feature Engineering

For each station, the last 20% of observed days form the test set, the most recent 10% of the remainder form the validation set and the earliest 70% form the training set (18,641 / 2,664 / 5,328 days). Twenty-nine base features are engineered per station: PM2.5 lags at 1/3/7/14/30 days, rolling means over 3/7/14 days and rolling standard deviations over 7/14 days, calendar variables, eight meteorological inputs and seven co-pollutants.

### C. Prophet Baseline and Residual-AR Augmentation

A separate Prophet model is fit per station on training-plus-validation observed PM2.5 with weekly and yearly seasonality and a changepoint prior scale of 0.05. Stations with fewer than sixty observations fall back to a seasonal-naive mean. The residual $r_t = y_t - \hat{y}_t^P$ is large and heavy-tailed (mean 2.0, standard deviation 43.4, range $[-199.5, 291.5]$ $\mu$g/m$^3$), confirming that Prophet leaves substantial structured signal for a neural learner.

Because the residual is itself temporally autocorrelated. Its recent history is appended to every learner: residual lags at 1/3/7 days, its 7-day rolling mean and standard deviation and Prophet's current and one-step-lagged output, expanding the feature representation from 29 to 36. All augmented features use at least a one-day shift.

### D. Proposed Model

Algorithm 1 presents the workflow of the proposed STAR model. Prophet first generates the baseline prediction and residual series for each monitoring station. Residual-based features are then organized into 14-day windows and processed using a Transformer encoder with station-aware FiLM conditioning. The network predicts the residual component, which is finally added to the Prophet baseline to produce the final $PM_{2.5}$ forecast.

**Algorithm 1** STAR: Station-Aware Transformer for Residual $PM_{2.5}$ Forecasting

**Require:** Daily $PM_{2.5}$ observations $X$, station IDs $S$
**Ensure:** One-day-ahead $PM_{2.5}$ prediction

**for** each monitoring station **do**
  Train a Prophet model using the training data
  Predict baseline $PM_{2.5}$ values
  Compute residuals:

$$R = Y - \hat{Y}_{Prophet}$$

**end for**
Generate residual features:
  Residual lags (1, 3, 7)
  Rolling mean and rolling standard deviation
  Prophet prediction and lagged prediction
Construct a 14-day sliding window for each sample
**for** each input window **do**
  Project input features into the Transformer embedding space
  Apply positional encoding
  Encode temporal dependencies using a Transformer encoder
  Average-pool temporal representations
  Obtain station embedding
  Generate FiLM parameters $(\gamma, \beta)$
  Apply station-conditioned feature modulation:

$$h = (1 + \gamma) \odot h + \beta$$

  Predict residual $\hat{R}$
**end for**
Compute the final prediction:

$$\hat{Y} = \hat{Y}_{Prophet} + \hat{R}$$

**return** $\hat{Y}$

### E. Residual Learners

Prophet+LSTM a single-layer 64-unit LSTM ingesting the same 14-day, 36-feature window with a small dropout head regressing the residual.

Prophet+ASRL a 128→64→32→1 MLP that concatenates an 8-dimensional station embedding to the features. Two ablations isolate design choices: a Generic MLP (no embedding, no batch normalisation) and ASRL (no emb.) (full architecture,

no embedding). The full ASRL is deep-ensembled over three seeds.

Prophet+STAR (proposed): each time step of the 14 day window is projected to a 64 dimensional space with a learned positional encoding and one Transformer encoder layer (four heads, feed-forward width 128) is applied. The encoded sequence is mean pooled to a vector $\mathbf{h}$, after which a station embedding $\mathbf{e}_s$ drives a FiLM modulation:

$$(\boldsymbol{\gamma}_s, \boldsymbol{\beta}_s) = W_{\text{FiLM}}\, \mathbf{e}_s, \quad \mathbf{h} \leftarrow (1 + \boldsymbol{\gamma}_s) \odot \mathbf{h} + \boldsymbol{\beta}_s, \qquad (2)$$

so that each station applies a per-channel scale and shift to the shared temporal representation. A layer normalisation and a small MLP head then output $\hat{r}$. STAR is deep-ensembled over three seeds. Unlike the additive concatenated embedding in ASRL, the multiplicative FiLM interaction lets a station gate which temporal features are emphasised.

### F. Training Protocol

All neural learners share one protocol: AdamW (lr $10^{-3}$, weight decay $10^{-2}$), Huber loss ($\delta = 1$) on the scaled residual, ReduceLROnPlateau scheduling, gradient-norm clipping at 5.0, batch size 32, up to 100 epochs and early stopping on validation loss with patience 10.

### G. Unified Statistical Evaluation

Every model is evaluated on a common test subset of 4,867 observed test days for which the windowed models can form a complete 14-day input. Performance is assessed using MAE, RMSE, $R^2$, MAPE, sMAPE and explained variance, each reported with 95% bootstrap confidence intervals obtained from 2,000 paired resamples.

Holm-corrected pairwise Wilcoxon tests are performed on per-sample absolute errors to identify statistically significant differences between model pairs. The adjusted p-values are visualized in Fig. 2. Nearly all model pairs exhibit statistically significant differences ($p < 0.05$), while the two ASRL variants show no significant difference, indicating that adding a static station embedding alone provides little improvement.

A Friedman omnibus test is then conducted on per-station MAE, treating the sixteen monitoring stations as statistical blocks. The resulting average model rankings are summarized in Table II. Prophet+STAR achieves the best overall average rank, followed by Prophet+LSTM, whereas the standalone Prophet baseline ranks last, demonstrating the overall effectiveness of the proposed residual-learning framework.

TABLE II
AVERAGE MODEL RANKINGS OBTAINED FROM THE FRIEDMAN TEST. LOWER AVERAGE RANK INDICATES BETTER OVERALL PERFORMANCE.

| Model | Average Rank | Mean MAE |
|---|---|---|
| Prophet+STAR (Proposed) | **1.667** | 18.528 |
| Prophet+LSTM [2] | 2.133 | 18.198 |
| Prophet+Generic MLP | 3.400 | 19.853 |
| Prophet+ASRL (no emb.) | 3.867 | 21.181 |
| Prophet+ASRL (emb.) | 3.933 | 21.611 |
| Prophet | 6.000 | 37.298 |

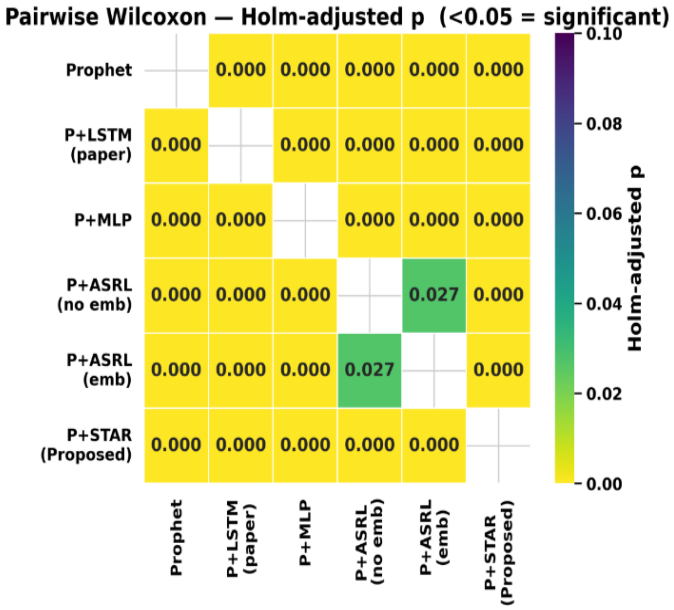



Fig. 2. Heatmap of Holm-adjusted pairwise Wilcoxon test p-values for all forecasting models.

## IV. Results and Analysis

### A. Overall Accuracy

Table III summarizes the forecasting performance of the proposed Prophet–STAR framework. The model achieved an MAE of 16.70 $\mu$g/m$^3$ and an RMSE of 27.20 $\mu$g/m$^3$, indicating good prediction accuracy. It obtained an $R^2$ score of 0.805 and an EVS of 0.809, showing that the model effectively explains the variability in the observed $PM_{2.5}$ concentrations. The MAPE and sMAPE values are 33.34% and 28.61% respectively, demonstrating stable forecasting performance across different pollution levels. The residual statistics further show a mean error of 4.32 $\mu$g/m$^3$ with a standard deviation of 26.85 $\mu$g/m$^3$. Although the maximum error reaches 223.29 $\mu$g/m$^3$, it is associated with a few extreme observations and does not substantially affect the overall prediction performance. Overall, the proposed Prophet–STAR framework provides accurate and reliable one-day-ahead $PM_{2.5}$ forecasts.

TABLE III
PERFORMANCE SUMMARY OF THE PROPOSED PROPHET+STAR MODEL

| **Forecasting Performance** | |
|---|---|
| **Metric** | **Value** |
| MAE | 16.71 |
| RMSE | 27.20 |
| $R^2$ | 0.805 |
| MAPE (%) | 33.34 |
| sMAPE (%) | 28.61 |
| EVS | 0.809 |
| **Residual Statistics** | |
| Mean Error | 4.32 |
| Std. Deviation | 26.85 |
| Maximum Error | 223.29 |

### B. Ablation Insights

The ablation study presented in Table IV evaluates the contribution of different residual learning architectures to the proposed forecasting framework. Prophet is used as the baseline model, while progressively more expressive residual learners are incorporated to assess their effectiveness in modeling the nonlinear prediction errors.

As shown in Table IV, the standalone Prophet model produces the weakest performance, with an MAE of 35.01 $\mu$g/m$^3$, an RMSE of 49.69 $\mu$g/m$^3$ and an $R^2$ score of only 0.347. These results indicate that although Prophet effectively captures long-term trend and seasonality, it is insufficient for modeling the complex nonlinear relationships present in $PM_{2.5}$ time-series data.

Replacing the residual learner with a Generic MLP substantially improves forecasting accuracy, reducing the MAE to 19.31 $\mu$g/m$^3$ and the RMSE to 29.79 $\mu$g/m$^3$, while increasing the $R^2$ score to 0.766. This demonstrates that learning the residual errors significantly enhances the predictive capability beyond the Prophet baseline.

To investigate the effect of the proposed Adaptive Station-aware Residual Learning (ASRL) architecture, two variants were evaluated. The ASRL model without station embeddings achieved an MAE of 20.09 $\mu$g/m$^3$ and an RMSE of 30.41 $\mu$g/m$^3$, whereas the version with station embeddings obtained an MAE of 20.30 $\mu$g/m$^3$ and an RMSE of 31.06 $\mu$g/m$^3$. The similar performance of these two variants indicates that incorporating static station embeddings alone does not provide a meaningful improvement in forecasting accuracy for the considered dataset.

The Prophet+LSTM model further improves prediction performance by effectively modeling sequential dependencies in the residual series. It achieves the lowest RMSE (27.09 $\mu$g/m$^3$), the highest $R^2$ (0.806) and the highest EVS (0.814), demonstrating the effectiveness of recurrent neural networks for residual forecasting.

Finally, the proposed Prophet+STAR framework achieves the best overall balance across the evaluation metrics. As reported in Table IV, it obtains the lowest MAE of 16.71 $\mu$g/m$^3$ and the lowest sMAPE of 28.61%, while maintaining an RMSE of 27.20 $\mu$g/m$^3$, an $R^2$ score of 0.805, a MAPE of 33.34% and an EVS of 0.809. Although its RMSE, $R^2$ and EVS are marginally lower than those of the Prophet+LSTM model, the differences are very small (0.11 $\mu$g/m$^3$ in RMSE and 0.001 in $R^2$). In contrast, the proposed model achieves the lowest MAE and sMAPE, indicating superior average prediction accuracy and more balanced relative errors.

Overall, the ablation results demonstrate that the STAR residual learning module provides the most effective trade-off between absolute prediction accuracy and robustness. The consistent improvements over the Generic MLP and both ASRL variants confirm that the proposed STAR architecture captures nonlinear residual patterns more effectively, thereby enhancing the overall forecasting capability of the hybrid Prophet-based framework.

TABLE IV
FORECASTING ACCURACY ON THE COMMON TEST SUBSET

| Model | MAE | RMSE | $R^2$ | MAPE | sMAPE | EVS |
|---|---|---|---|---|---|---|
| Prophet | 35.01 | 49.69 | 0.347 | 71.76 | 55.08 | 0.354 |
| Prophet+Generic MLP | 19.31 | 29.79 | 0.766 | 38.50 | 30.68 | 0.766 |
| Prophet+ASRL (no emb.) | 20.09 | 30.41 | 0.756 | 38.53 | 32.03 | 0.759 |
| Prophet+ASRL (emb.) | 20.30 | 31.06 | 0.745 | 40.45 | 33.53 | 0.752 |
| Prophet+LSTM [2] | 17.15 | **27.09** | **0.806** | **33.08** | 29.78 | **0.814** |
| Prophet+STAR (proposed) | **16.71** | 27.20 | 0.805 | 33.34 | **28.61** | 0.809 |

### C. Model Convergence Analysis

Figure 3 illustrates the training and validation Huber loss curves of the proposed Prophet+STAR model. The model converges rapidly during the initial epochs, followed by a gradual and stable decrease in both training and validation losses. Throughout the training process, the two curves remain closely aligned with only a small and consistent gap, indicating stable optimization and good generalization capability. Furthermore, the absence of a sustained increase in the validation loss suggests that the proposed model does not suffer from significant overfitting. These observations demonstrate the effectiveness of the adopted training strategy in achieving robust convergence while maintaining strong predictive performance.

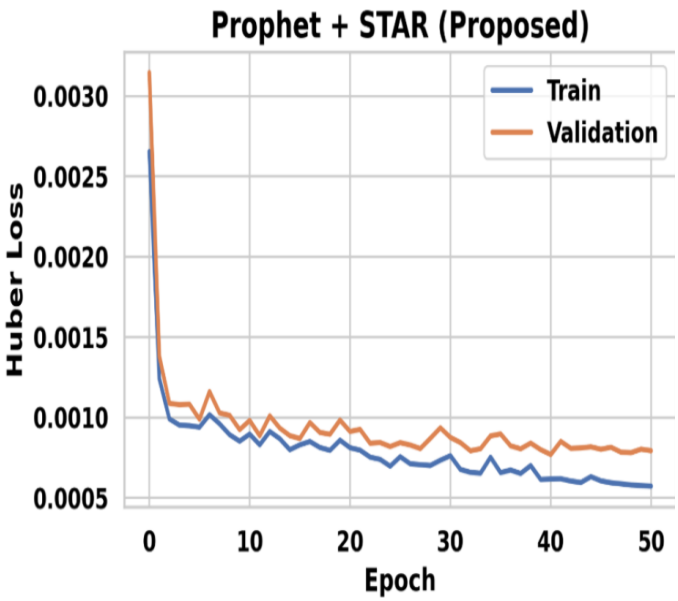



Fig. 3. Training and validation loss curves of Prophet+LSTM and the proposed STAR model.

## V. CONCLUSION

This paper introduced STAR, a station-aware temporal attention residual network for multi-station PM2.5 forecasting and evaluated it against a reproduced Prophet+LSTM, three feed-forward residual baselines and Prophet alone on a nineyear, sixteen-station Bangladesh dataset. Rather than relying on a single split or a lone pairwise test, all models were assessed on one common test subset within a unified statistical

framework bootstrap confidence intervals, a Friedman omnibus test, Holmcorrected Wilcoxon tests and a Nemenyi critical-difference analysis with stations as blocks. STAR achieved the best MAE and symmetric MAPE, statistically matched the recurrent baseline under conservative criteria while holding a per-sample significant edge and decisively outperformed every feedforward residual learner. The experiments show that temporal modeling of the Prophet residual is the decisive factor, that a static station embedding adds little on its own and that attention combined with FiLM station conditioning attains recurrent-level accuracy at lower architectural cost with added interpretability and explicit station awareness. These findings support STAR as a lightweight, defensible alternative to recurrent residual learners for environmental time-series forecasting.Future work will extend STAR by incorporating graph-based spatial relationships among monitoring stations and additional meteorological information. Its effectiveness will also be validated on larger multi-region air-quality datasets to further assess robustness and generalizability.

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